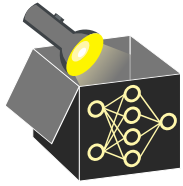
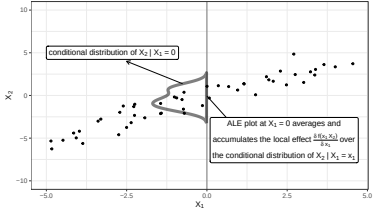
Interpretable Machine Learning



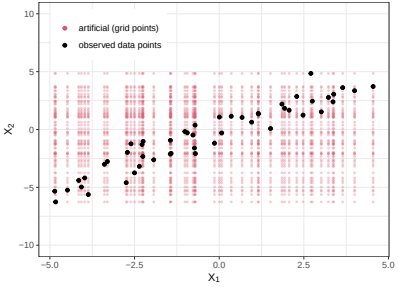
Feature Effects
Accumulated Local Effect (ALE): Intro

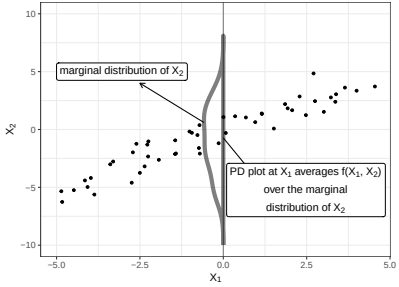


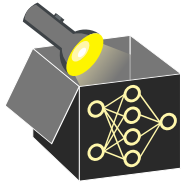
Learning goals

- PD plots and its extrapolation issue
- M plots and its omitted-variable bias
- Understand ALE plots

MOTIVATION - CORRELATED FEATURES

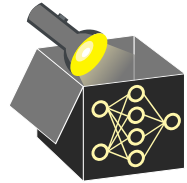






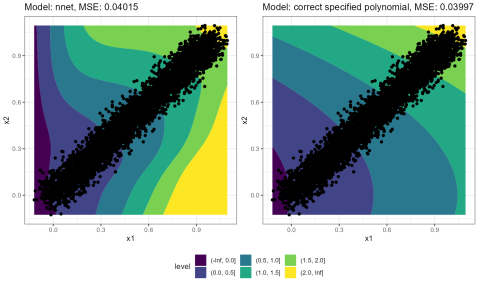
- PD plots **average over predictions** of artificial points that are out of distribution/ unlikely (red)
  $\Rightarrow$ Can lead to misleading / biased interpretations, especially if model also contains interactions
- Not wanted if interest is to interpret effects within data distribution

MOTIVATION - CORRELATED FEATURES



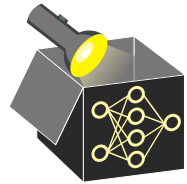
Example: Fit an NN to 5000 simulated data points with $x \sim Unif(0, 1)$, $\epsilon \sim N(0, 0.2)$ and

$$y = x_1 + x_2^2 + \epsilon, \text{ where } x_1 = x + \epsilon_1,\ x_2 = x + \epsilon_2 \text{ and } \epsilon_1, \epsilon_2 \sim N(0, 0.05).$$



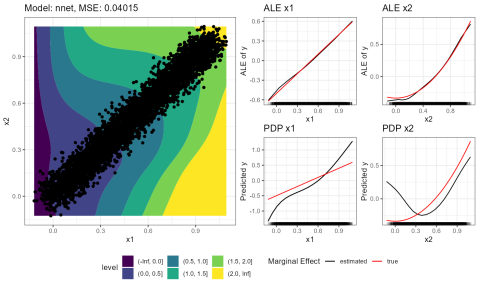
- Test error (MSE) of NN is comparable to other models
- NN contains interactions (see complex pred. surface)

MOTIVATION - CORRELATED FEATURES



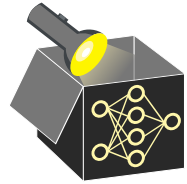
Example: Fit an NN to 5000 simulated data points with $x \sim Unif(0, 1)$, $\epsilon \sim N(0, 0.2)$ and

$$y = x_1 + x_2^2 + \epsilon, \text{ where } x_1 = x + \epsilon_1,\ x_2 = x + \epsilon_2 \text{ and } \epsilon_1, \epsilon_2 \sim N(0, 0.05).$$

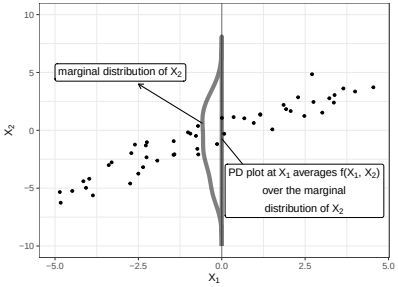


- Test error (MSE) of NN is comparable to other models
- NN contains interactions (see complex pred. surface)
- ALE in line with ground truth
- PDP does not reflect ground truth effects of DGP well
  $\Rightarrow$ Due to interactions and averaging of points outside data distribution

M PLOT VS. PD PLOT

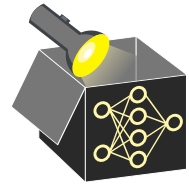


a) PD plot

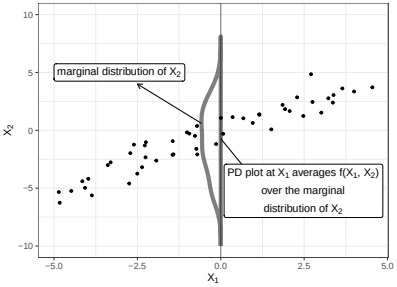


a) PD plot $\mathbb{E}_{\boldsymbol{x}_2}(\hat{f}(x_1, \boldsymbol{x}_2))$ is estimated by $\hat{f}_{1,PD}(x_1) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(x_1, \boldsymbol{x}_2^{(i)})$

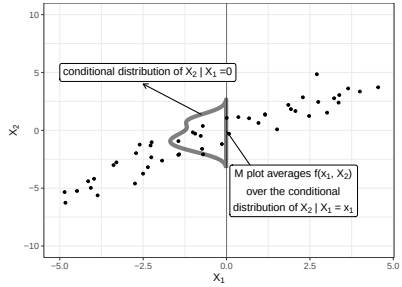
M PLOT VS. PD PLOT



a) PD plot



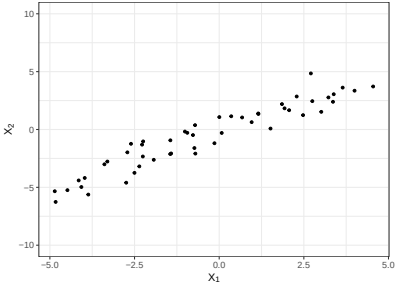
b) M plot

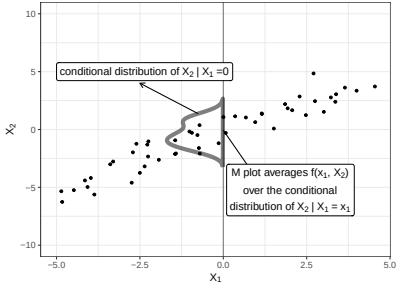


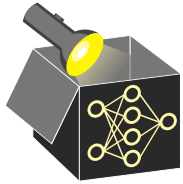
a) PD plot $\mathbb{E}_{\boldsymbol{x}_2}\big(\hat{f}(x_1, \boldsymbol{x}_2)\big)$ is estimated by $\hat{f}_{1,PD}(x_1) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(x_1, \boldsymbol{x}_2^{(i)})$

b) M plot $\mathbb{E}_{\boldsymbol{x}_2|\boldsymbol{x}_1}\big(\hat{f}(x_1, \boldsymbol{x}_2) \mid \boldsymbol{x}_1\big)$ is estimated by
$\hat{f}_{1,M}(x_1) = \frac{1}{|N(x_1)|}\sum_{i \in N(x_1)} \hat{f}(x_1, \boldsymbol{x}_2^{(i)})$, where index set
$N(x_1) = \{i : x_1^{(i)} \in [x_1 - \epsilon, x_1 + \epsilon]\}$ refers to observations with feature value close to x_1.

M PLOT VS. PD PLOT

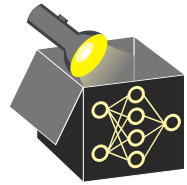


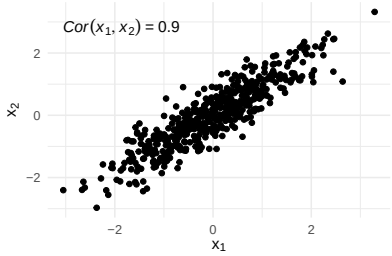




- M plots average predictions over conditional distribution (e.g., $\mathbb{P}(\boldsymbol{x}_2|x_1)$)
  $\Rightarrow$ Averaging predictions close to data distrib. avoids extrapolation issues
- **But:** M plots suffer from omitted-variable bias (OVB)
  - Because of the conditioning M plots contain effects of other dependent features
  - Useless in assessing a feature's marginal effect if feature dependencies are present

M PLOT VS. PD PLOT - OVB EXAMPLE





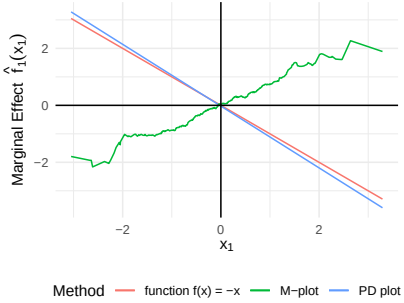
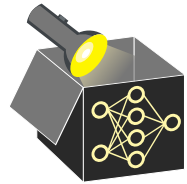


Illustration: Fit LM on 500 i.i.d. observations with features $x_1, x_2 \sim N(0, 1)$, $Cor(x_1, x_2) = 0.9$ and

$$y = -x_1 + 2 \cdot x_2 + \epsilon \quad \epsilon \sim N(0, 1)$$

Results: M plot of x_1 also includes marginal effect of all other dependent features (here: x_2)

IDEA: INTEGRATING PARTIAL DERIVATIVES

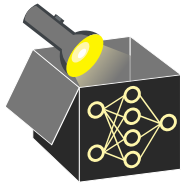


Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\boldsymbol{x}_j$ removes other main effects

$\Rightarrow$ Integrating again w.r.t. $\boldsymbol{x}_j$ recovers the original main effect of $\boldsymbol{x}_j$

IDEA: INTEGRATING PARTIAL DERIVATIVES



Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\boldsymbol{x}_j$ removes other main effects

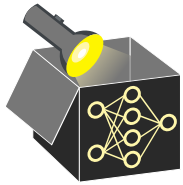
$\Rightarrow$ Integrating again w.r.t. $\boldsymbol{x}_j$ recovers the original main effect of $\boldsymbol{x}_j$

Example:

- Consider an additive prediction function:

$$\hat{f}(x_1, x_2) = 2x_1 + 2x_2 - 4x_1x_2$$

IDEA: INTEGRATING PARTIAL DERIVATIVES



Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\boldsymbol{x}_j$ removes other main effects

$\Rightarrow$ Integrating again w.r.t. $\boldsymbol{x}_j$ recovers the original main effect of $\boldsymbol{x}_j$

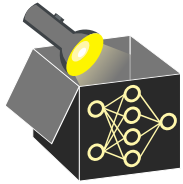
Example:

- Consider an additive prediction function:

$$\hat{f}(x_1, x_2) = 2x_1 + 2x_2 - 4x_1x_2$$

- Partial derivative of $\hat{f}$ w.r.t. x_1: $\frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} = 2 - 4x_2$

IDEA: INTEGRATING PARTIAL DERIVATIVES



Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\boldsymbol{x}_j$ removes other main effects

$\Rightarrow$ Integrating again w.r.t. $\boldsymbol{x}_j$ recovers the original main effect of $\boldsymbol{x}_j$

Example:

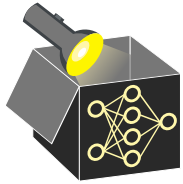
- Consider an additive prediction function:

$$\hat{f}(x_1, x_2) = 2x_1 + 2x_2 - 4x_1x_2$$

- Partial derivative of $\hat{f}$ w.r.t. x_1: $\frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} = 2 - 4x_2$
- Integral of partial derivative ($z_0 = \min(x_1)$):

$$\int_{z_0}^{x} \frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} dx_1 = [2x_1 - 4x_1x_2]_{z_0}^{x}$$

IDEA: INTEGRATING PARTIAL DERIVATIVES



Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\boldsymbol{x}_j$ removes other main effects

$\Rightarrow$ Integrating again w.r.t. $\boldsymbol{x}_j$ recovers the original main effect of $\boldsymbol{x}_j$

Example:

- Consider an additive prediction function:

$$\hat{f}(x_1, x_2) = 2x_1 + 2x_2 - 4x_1x_2$$

- Partial derivative of $\hat{f}$ w.r.t. x_1: $\frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} = 2 - 4x_2$
- Integral of partial derivative ($z_0 = \min(x_1)$):

$$\int_{z_0}^{x} \frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} dx_1 = [2x_1 - 4x_1x_2]_{z_0}^{x}$$

- We removed the main effect of x_2, which was our goal